An Investigation into the Use of Numerical Methods to Find Eigenvectors and Eigenvalues in Quantum Mechanics

Katie Heath
School of Physics, University of Bristol.
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PART ONE

Method

In order to find the eigenvectors and eigenvalues of the quantum harmonic oscillator, a function was written which uses power iteration and hotelling deflation to return an array of the eigenvectors and an array of the eigenvalues associated with the Haamiltonian matrix passed to the function. This approach was used for two reasons: firstly, to advance my Python skills by coding new numerical methods taught in this course, and secondly to explore how this compared to the latter used built in SciPy functions.

Power iteration was used to find the eigenvectors of the quantum harmonic oscillator (QHO) at each discrete point N, the Rayleigh Quotient was then used to find the eigenvalue associated with this eigenvector, followed by the use of hotelling deflation to extract each identified eigensolution from the Hamiltonian matrix so that the function produces a new eigensolution with each iteration of the function. [1]

Results and Discussion

The Hamiltonian was validated through a function written to calculate both sides of the eigenvalue equation, namely $H\psi = E\psi$ where H is the Hamiltonian matrix, E are the eigenvalues and ψ are the eigenvectors. This was used in this section of the project primarily to confirm that the self-written numerical method function was working as expected.

Table I shows the energy levels obtained from this code in comparison to the expected energy levels of the QHO [2]. From this it can be seen that a reasonable level of accuracy was achieved for the lower energy levels, however for each increasing energy level the percentage error increased by about 0.6% from the previous energy level. It can thus be concluded that this numerical method is unreliable for computing the energy levels of QHO systems with high energy levels. This is likely because of the nature of the Hamiltonian being a large, sparsely populated matrix with many similar values, which makes hotelling deflation a less reliable process. [3]

This is also supported by the numerical function, which attempts to have iterations converge up to eight decimal places before adding a result to the array. However, as the value of N is increased, the more eigenvectors it fails to do this for as it stops after 50000 iterations to prevent infinite loops occurring. This means that some of the resulting eigensolutions do not fit the desired level of precision and further supports that this method is not ideal for this purpose.

Energy level n	Produced value ($\hbar\omega$)	Expected value ($\hbar\omega$)	% error
0	0.4996	0.5000	0.0638
1	1.4984	1.5000	0.1064
2	2.4958	2.5000	0.1661
3	3.4920	3.5000	0.2283
4	4.4868	4.5000	0.2912

TABLE I: A comparison of the eigenvalues produced by numerical methods and the expected eigenvalues [2] for the QHO.

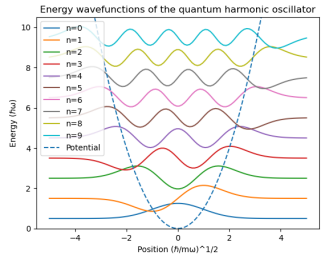


FIG. 1: A plot of the first 10 wavefunctions of the QHO, with a plot of the potential imposed on top.

Additionally, the results were qualitatively analysed by comparing the produced plots with those from the scientific literature. From Figures 1, 2 and 3, it can be seen that the wavefunctions produced by the code produced visibly accurate results when compared with the expected results. This is to be expected due to the small percentage errors shown in Table I

Although this numerical method produced accurate plots, it did so at a very slow pace, which further sug-

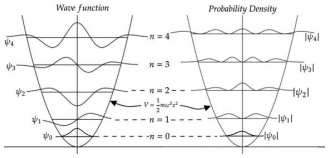
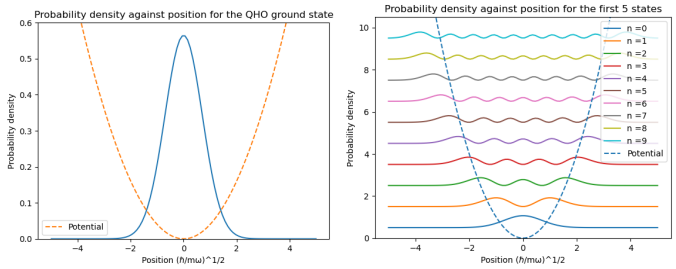


FIG. 2: An example plot of the first 5 wavefunctions and their probability densities of the QHO [4].



(a) A plot of the probability density of the ground state of the QHO

(b) A plot of the probability density of the first 3 states of the QHO.

FIG. 3: The probability density of the QHO.

gests that this may not be the most effective method to approach this problem, as the long processing time resulted in needing to use a lower value of N to achieve faster results, which lowers the accuracy of the function.

PART TWO

Method

To find the eigenvectors and eigenvalues of the hydrogen atom, the function written for the first part of the project was initially used. This produced results very slowly and produced wavefunctions which rapidly oscillated between positive and negative values, producing graphs which appeared as though the wavefunction of the hydrogen atom was acting as an envelope function for second sinusoidally oscillating function. Instead, a function from the SciPy library, scipy.sparse.linalg.eigsh, was used to produce a more efficient and more accurate code. This specific function was chosen as it is designed for sparsely populated, large arrays, as is the nature of the Hamiltonian in this problem, especially as larger values of N are used to further improve the precision of the code[5][6]. Additionally, further efficiency was required for this section of the project as the eigensolutions were calculated for 5 different values of the quantum number l, in order to model the state of the hydrogen atom under a range of circumstances.[7]

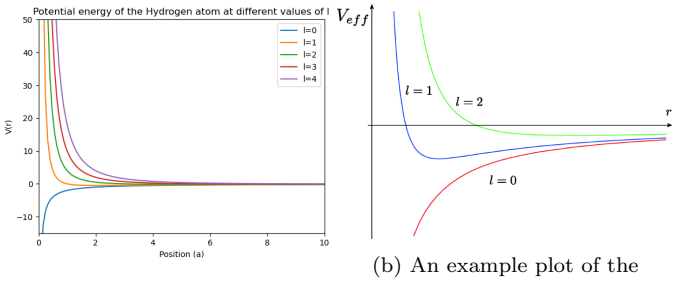
Henceforth, the code solves this problem by using a built in function from the SciPy library to find the eigensolutions, and then proceeds to verify these results through comparison of the quantisation of the produced energy levels, the wavefunction, and the probability density of the hydrogen atoms at 5 different values of l.

Results and Discussion

For this section of the code, the Hamiltonian matrix was validated by multiplying it by its transpose to check it was hermitian as expected. The potential matrix was validated through a plot of the potentials, shown in Figure 4. Both of these validations were positive.

Energy level (n)	Produced value	Expected value	% error
1	-0.9996	-1.0000	0.0399
2	-0.2499	-0.2500	0.0006
3	-0.1111	-0.1111	0.0001
4	-0.0611	-0.0625	0.0086
5	-0.0222	-0.0400	0.0711

TABLE II: A comparison of the eigenvalues produced by numerical methods and the expected eigenvalues [7] for the hydrogen atom when $l = 0$, with all energy levels in units ($\frac{m_e e^4}{2(4\pi\epsilon_0)^2\hbar^2}$).



(a) A plot of the potential energy of the hydrogen atom at 5 different values of l.

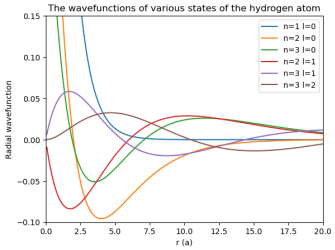
(b) An example plot of the expected potential energy of the hydrogen atom at 3 different values of l. [8]

FIG. 4: The potential energy of the hydrogen atom.

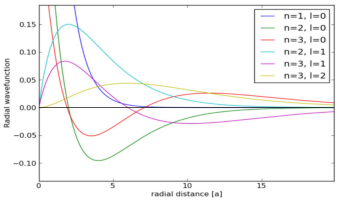
From Table II it can immediately be noted that the percentage errors, and hence eigenvalues, produced by the built in SciPy function are roughly an order of magnitude larger than those produced in part one. This suggests that this numerical method is more precise and accurate, as well as faster, than the other.

A comparison of the plots produced and the expected plots for the wavefunctions and probability densities of various hydrogen atom states can be seen in Figures 5, 6 and 7. The similarity of these further supports the claim that the numerical method provided by SciPy is the more accurate method.

However, it can be seen in Figure 5 that the $n = 2, l = 1$ line has been plotted upside down, a problem which can also be seen in Figure 1 from the first part of this project, although less noticable due to the nature of the function. This arises because both the positive and negative values of an eigenvector are valid results, and whether a positive or negative eigenvector is found depends on the initial estimates of the eigenvectors. Hence, this is not an error of either numerical method.

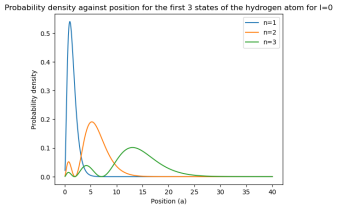


(a) A plot of the radial wavefunctions of the hydrogen atom when for a range of l values at a selection of energy levels.

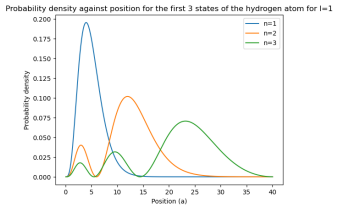


(b) An example plot of the same radial wavefunction of the hydrogen atom for comparison [4].

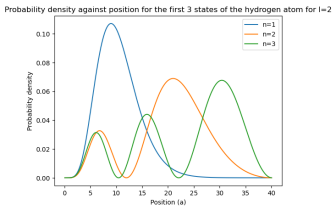
FIG. 5: The probability density of the first 5 wavefunctions of the hydrogen atom, at 4 different values of l.



(a) A plot of the probability density of the first 3 wavefunctions of the hydrogen atom when $l = 0$.



(b) A plot of the probability density of first 3 wavefunctions of the hydrogen atom when $l = 1$..



(c) A plot of the probability density of the first 3 wavefunctions of the hydrogen atom when $l = 2$.

FIG. 6: The probability density of the first 3 wavefunctions of the hydrogen atom, at 3 different values of l.

CONCLUSION

Creating my own function produced results much slower, resulting in evaluating the wavefunctions at less discrete points, creating less accurate continuous solutions. The use of hotelling deflation in this function means it naturally produces the largest eigenpairs first, however for a problem of this nature we are primarily interested in the smaller values, namely the ground state of the particle and the first few excited states, this meant having to calculate the eigenvalue for every value of N, rather than just those necessary. This is also a contributing factor in this numerical method taking longer than the SciPy function, which only finds the smallest states. Overall the built in function is a better fit for the purpose of this project

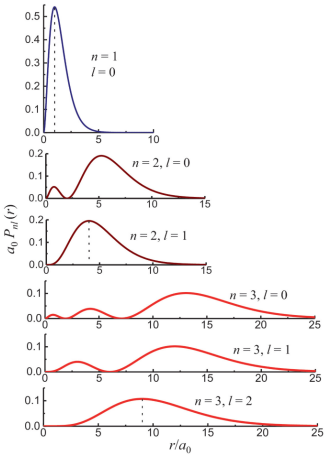


FIG. 7: Some example plots of probabiltiy densities of different states of the hydrogen atom.[9]

This project has developed my ability to efficiently write functions for numerical methods, thus increasing my confidence and experience in using Python. The comparison of using the two different methods taught me that built in functions are almost always the most efficient method of tackling problems in computational physics, a lesson which I will take forward to further develop my approach to future projects. Finally, I have also learnt how to use many new linear algebra functions from the SciPy library, which has developed my experience with this resource.

USE OF GENERATIVE AI

Generative AI was used in the creation of this code to identify where failures in the verification methods originated from when I could not figure this out alone. It was also used to provide suggested solutions to these errors when I could not find a working solution from researching the internet or my own prior coding knowledge. The only generative AI model used in this assignment was ChatGPT, and any use of this is referenced, with the exact prompt entered, in the code.

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